Sai Pranay Chebium

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SUMMARY

Full-Stack Developer and Computer Science graduate student with experience designing scalable web and mobile applications and building robust backend systems. Skilled in API architecture, distributed computing, and performance optimization, with published research in machine learning and graph-based algorithms. Passionate about developing reliable, maintainable software and contributing to high-impact engineering teams.

EDUCATION

M.S. Computer Science, Florida State University — GPA: 3.944 / 4.0 — 2024–2026
B.Tech Computer Science, SRM University - AP — GPA: 3.83 / 4.0 — 2019–2023

TECHNICAL SKILLS

Languages: Python, C++, Java, C, JavaScript, TypeScript, SQL
Frontend: React, Next.js, React Native, SCSS, HTML, CSS
Backend: Node.js, FastAPI, REST APIs, GraphQL, Prisma
Databases: MySQL, NoSQL, PostgreSQL, Firebase
Cloud & DevOps: AWS, Google Cloud, Azure DevOps, Docker, Kubernetes, Terraform, CI/CD
Tools: Git, Android Studio, Xcode, VS Code, Postman, LaTeX

EXPERIENCE

MAJD Digital — **Dubai, UAE (Remote)**
Full Stack Developer — Nov 2022–Dec 2023

- Engineered and deployed scalable web and mobile applications using React, React Native, Node.js, and Firebase, supporting 1,000+ active users.
- Designed and optimized 15+ RESTful API endpoints, improving backend response time by 35% and reducing crash rates by 43%.
- Refactored legacy modules and improved database query efficiency, decreasing technical debt and enhancing application stability across releases.
- Led end-to-end feature development in agile sprints, from requirement analysis to production deployment, ensuring timely delivery and high code quality.
- Implemented authentication workflows, role-based access control, and secure data handling practices to strengthen application security.
- Improved application scalability by restructuring backend services and optimizing asynchronous processing pipelines.
- Collaborated closely with UI/UX designers and product stakeholders to translate business requirements into production-ready features.

University Teknologi Petronas — **Malaysia**
Intern — Jun–Jul 2022

- Contributed to collaborative full-stack development projects, applying OOP principles and Git-based version control workflows.
- Assisted in system requirement analysis and rapid prototyping, accelerating feature validation cycles by 20%.
- Participated in design discussions and code reviews, strengthening understanding of modular system design and maintainable architecture.

RESEARCH

Software Maintainability Prediction Using Ensemble Learning — IEEE 2023
Developed ensemble ML models using 10+ code metrics, improving prediction accuracy by 18% over baseline classifiers. Conducted comparative evaluation across multiple classifiers and validated results using cross-validation techniques to ensure model robustness.

Distributed and Parallel Algorithms for Link Prediction — Under Review
Designed scalable distributed approaches for link prediction on large-scale social network graphs (100K+ nodes), reducing computation time by 35%. Analyzed graph partitioning strategies and optimized parallel computation to enhance performance across distributed environments.

Disease Prediction using Symptom Classification — Under Review
Built supervised classification models on structured medical datasets, achieving 90%+ validation accuracy for early-stage disease prediction. Performed feature selection, model tuning, and evaluation to improve generalization and reduce overfitting in clinical prediction tasks.

PROJECTS

Socio – Real-Time Chat Application
Built using Next.js, MongoDB, Prisma, and Pusher. Implemented authentication, group messaging, and real-time updates.

MerchWorld – E-Commerce Platform
Developed React + FastAPI marketplace with fraud-prevention mechanisms and scalable backend APIs.

NIFTY – Stock Market Analysis and Prediction System
Designed and implemented a financial data analysis model for stock trend prediction using historical market datasets. Applied data preprocessing, feature selection, and predictive modeling techniques to evaluate market movement patterns and improve decision-making accuracy.

MDUAL – Secure Dual Authentication System
Developed a multi-layer authentication framework integrating user credential validation and secondary verification mechanisms to enhance system security. Designed system workflows emphasizing secure access control and reliability.

Meble – Android Application
Java-based mobile app for furniture browsing and ordering with backend integration.